

Module 7: Molecular Genetics - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. In DNA, the base adenine always pairs with:

- A. cytosine
- B. guanine
- C. thymine
- D. another adenine
- Answer: C
- Why: The base-pairing rules are A with T and C with G. This complementarity lets DNA be copied accurately.

2. DNA replication is described as "semiconservative" because each new double helix contains:

- A. one original strand and one new strand
- B. two brand-new strands
- C. two original strands
- D. no strands at all
- Answer: A
- Why: Each daughter molecule keeps one parental strand as a template, so half of the original is conserved.

3. Which sequence shows the normal flow of genetic information?

- A. protein, then RNA, then DNA
- B. RNA, then DNA, then protein
- C. protein, then DNA, then RNA
- D. DNA, then RNA, then protein
- Answer: D
- Why: Transcription copies DNA into RNA, and translation reads codons to build protein: DNA to RNA to protein.

4. A mutation changes a single DNA base. Why might this matter?

- A. DNA cannot change, so nothing happens.
- B. It can change a codon, which may change an amino acid and the protein.
- C. It always improves the organism.
- D. It affects only RNA and never proteins.
- Answer: B
- Why: A base change can alter a codon and therefore the protein. Effects range from none to significant.